Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8

Week 8

Lecture 15

15.1 Grover's algorithm: Overview and intuition

15.2 The Oracle

15.3 Inversion about the mean

15.4 Example Quantum Search

Lecture 16

16.1 Grover's algorithm Example

16.2 Probability of success

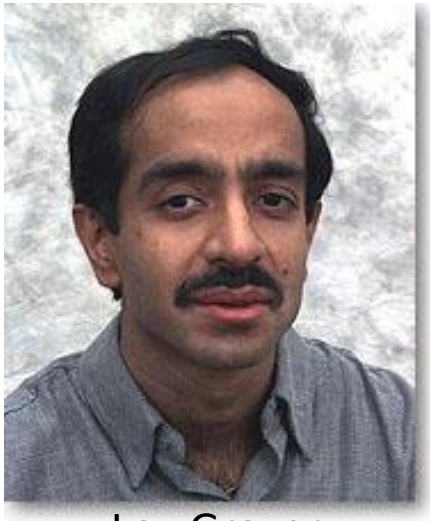
16.3 Geometric interpretation

Practice class 8

Grover's algorithm using QUI

Recap: Grover's Algorithm

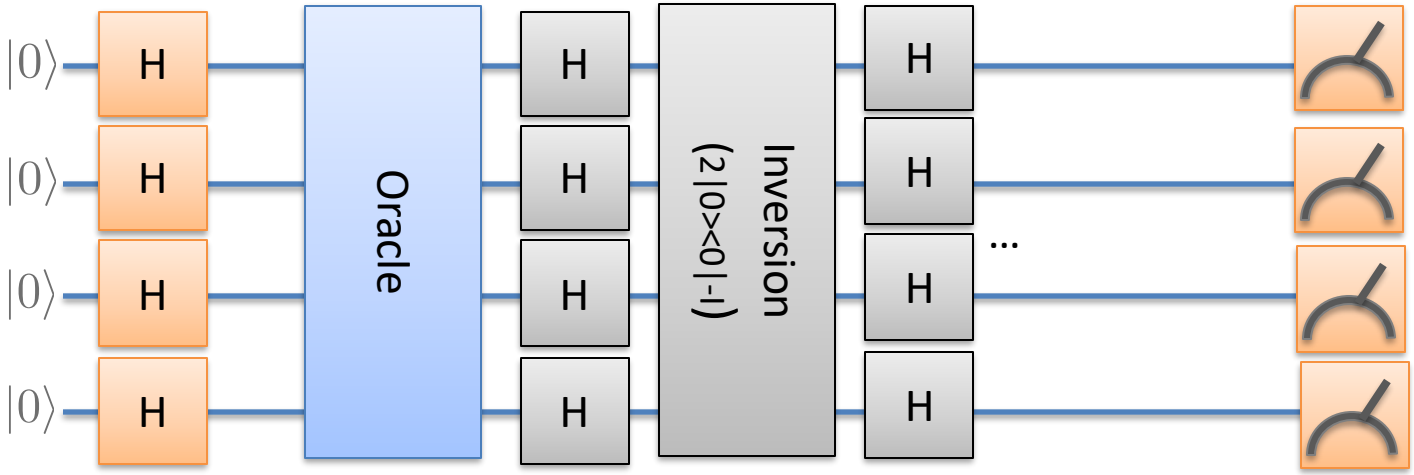
- Unordered search, find marked item among many
- Classically, this requires N/2 queries to the oracle
- Quantum mechanically, requires only O(√N) queries.



Lov Grover

Simple problem = search for one integer marked by the oracle.

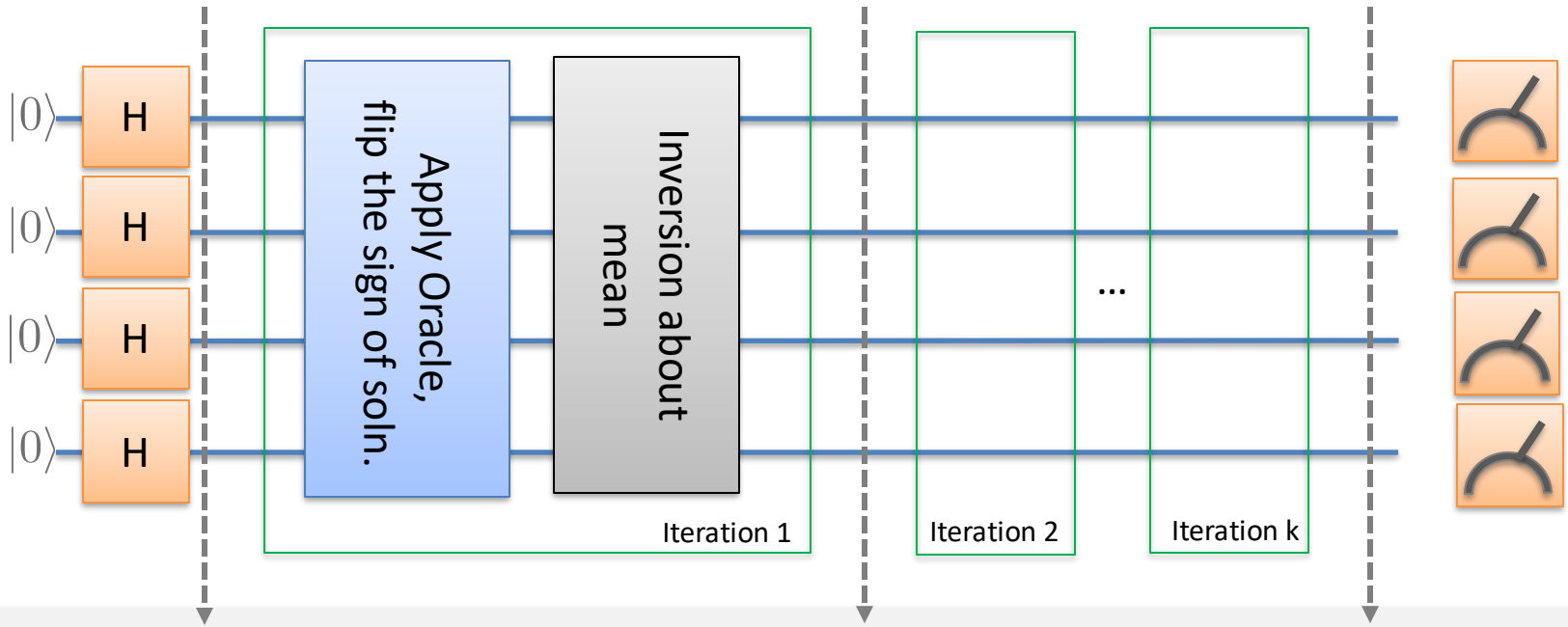
High level structure:



Recap: Grover's algorithm

Equal superposition

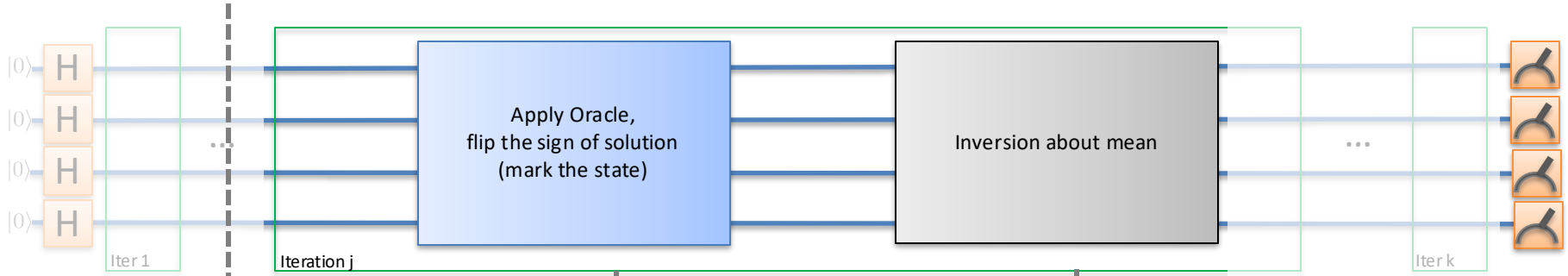
Solution has higher probability



Alice's coffee problem

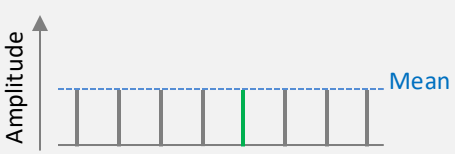
Intuition: Each iteration amplifies the amplitude of solution vectors.

Recap: Grover's algorithm

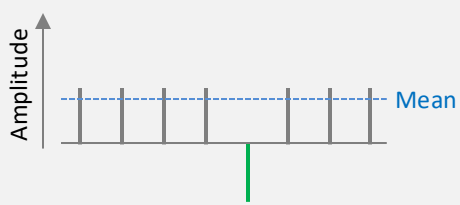


Maps every amplitude Mean$-\delta$ to Mean$+\delta$

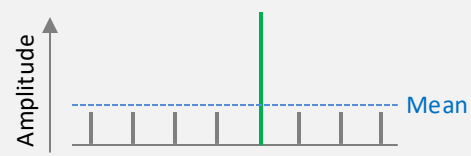
Alice's coffee problem



0 1 2 3 4 5 6 7



0 1 2 3 4 5 6 7



0 1 2 3 4 5 6 7

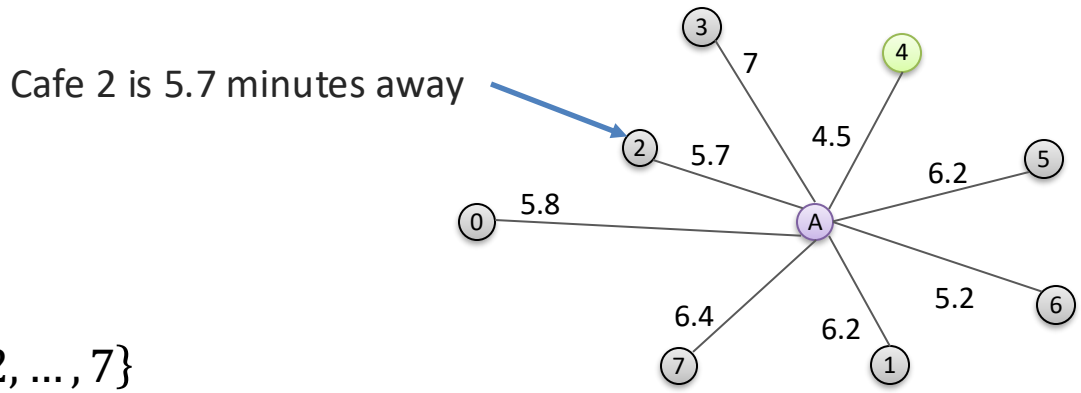
Intuition:

Flipping the sign of the amplitude of solution vector reduces the mean.

Inversion about the mean increases the amplitude of solution vector.

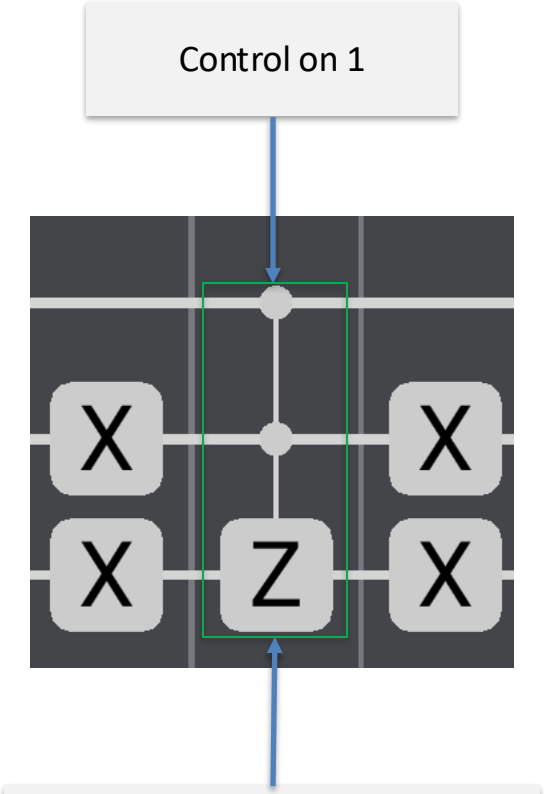
Unstructured search problems

- Example: Alice (A) wants to find a coffee shop within 5 mins of walking distance.



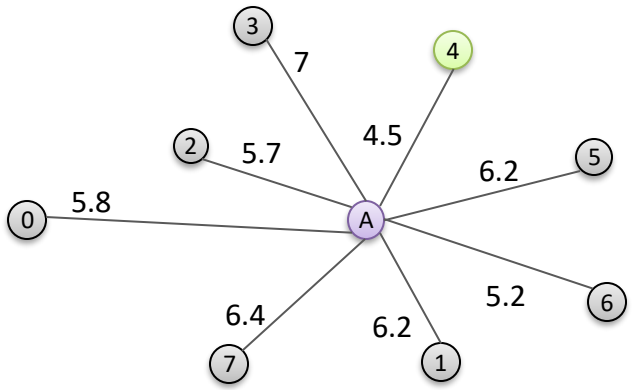
- $X = \{0,1,2,\dots,7\}$
  - Set X is not ordered by travel time
- $f(x) = \begin{cases} 1, \text{ if time(x)} \leq 5 \\ 0, \text{ otherwise} \end{cases}$
  - In general, the function f does not need to know about the search space.
- Solution x=4 (Cafe 4 is the only one that is less than 5 minutes away)
- The function f is like a blackbox (similar to Deutsch-Jozsa and Simon's algorithms).

The Oracle on QUI



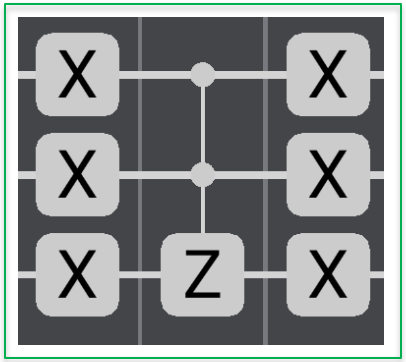
Flips the sign if:
- Control 1 qubit is in 1
- Target qubit is in 1

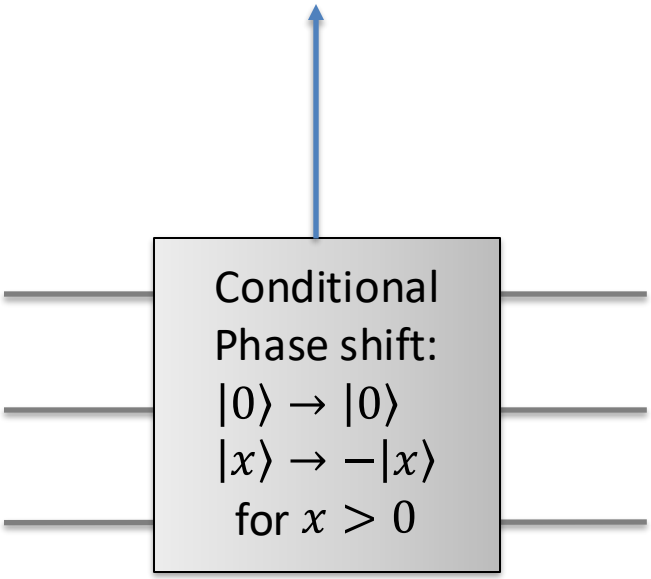
- $|000\rangle \rightarrow |000\rangle$
- $|001\rangle \rightarrow |001\rangle$
- $|010\rangle \rightarrow |010\rangle$
- $|011\rangle \rightarrow |011\rangle$
- $|100\rangle \rightarrow -|100\rangle$ (Solution state x=4)
- $|101\rangle \rightarrow |101\rangle$
- $|110\rangle \rightarrow |110\rangle$
- $|111\rangle \rightarrow |111\rangle$



Alice's coffee example.

Conditional Phase Shift on QUI





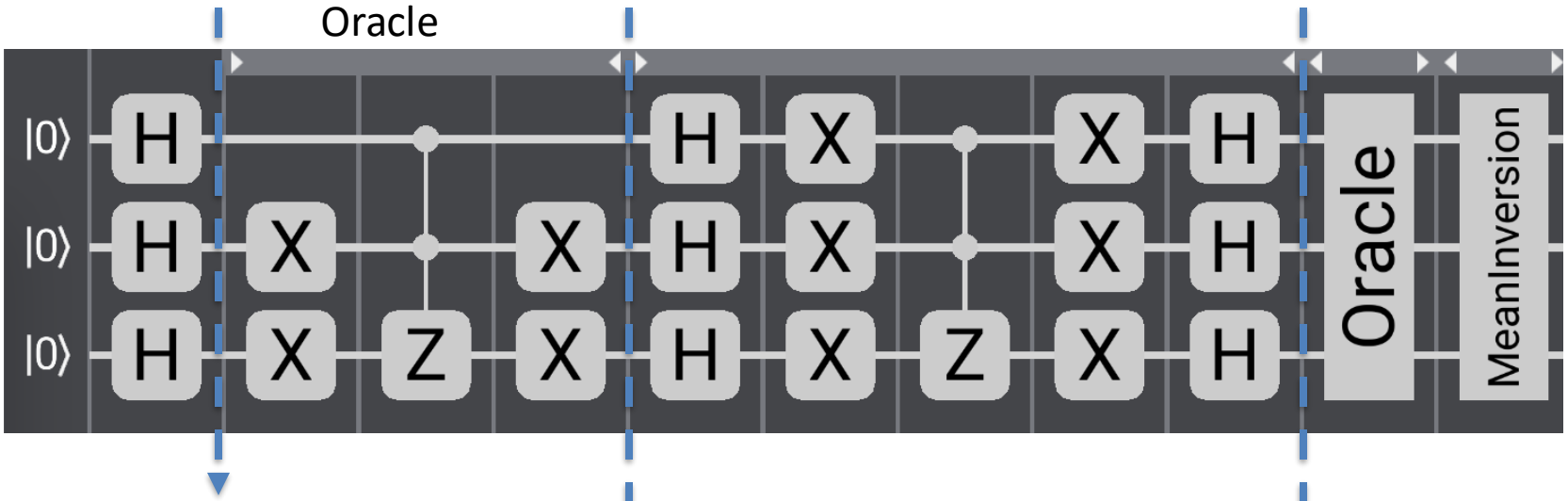
- $|000\rangle \to -|000\rangle$
- $|001\rangle \to |001\rangle$
- $|010\rangle \to |010\rangle$
- $|011\rangle \to |011\rangle$
- $|100\rangle \to |100\rangle$
- $|101\rangle \to |101\rangle$
- $|110\rangle \to |110\rangle$
- $|111\rangle \to |111\rangle$

Note: The QUI implementation does

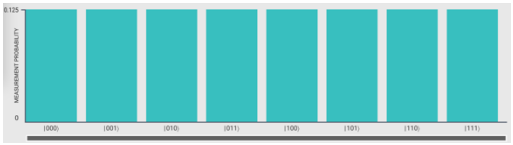
$$|0\rangle \to -|0\rangle$$
$$|x\rangle \to |x\rangle$$

This is correct as global phase does not affect probabilities of measuring a state.

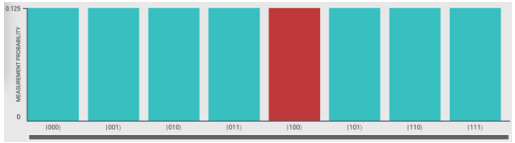
Solving Alice's coffee problem on QUI



Equal superposition

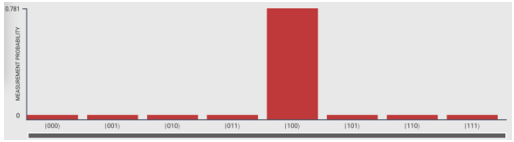


Marking of the solution state



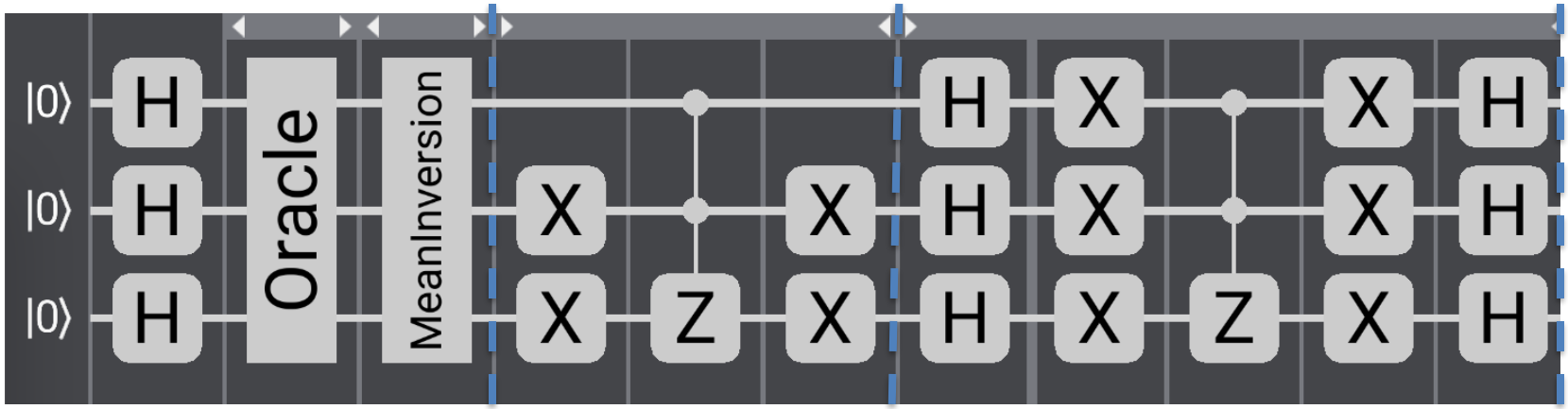
0.125

Mean inversion (1st iteration)

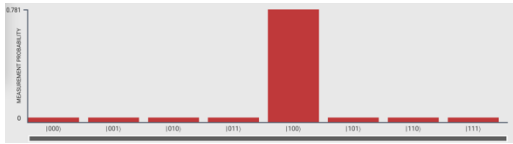


0.781

Solving Alice's coffee problem on QUI

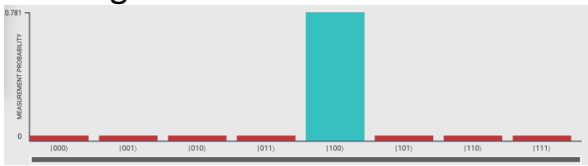


Probabilities after first iteration



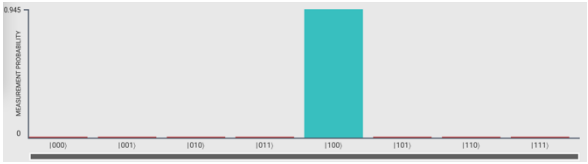
0.781

Marking of the solution state



0.781

Mean inversion (2nd iteration)



0.945

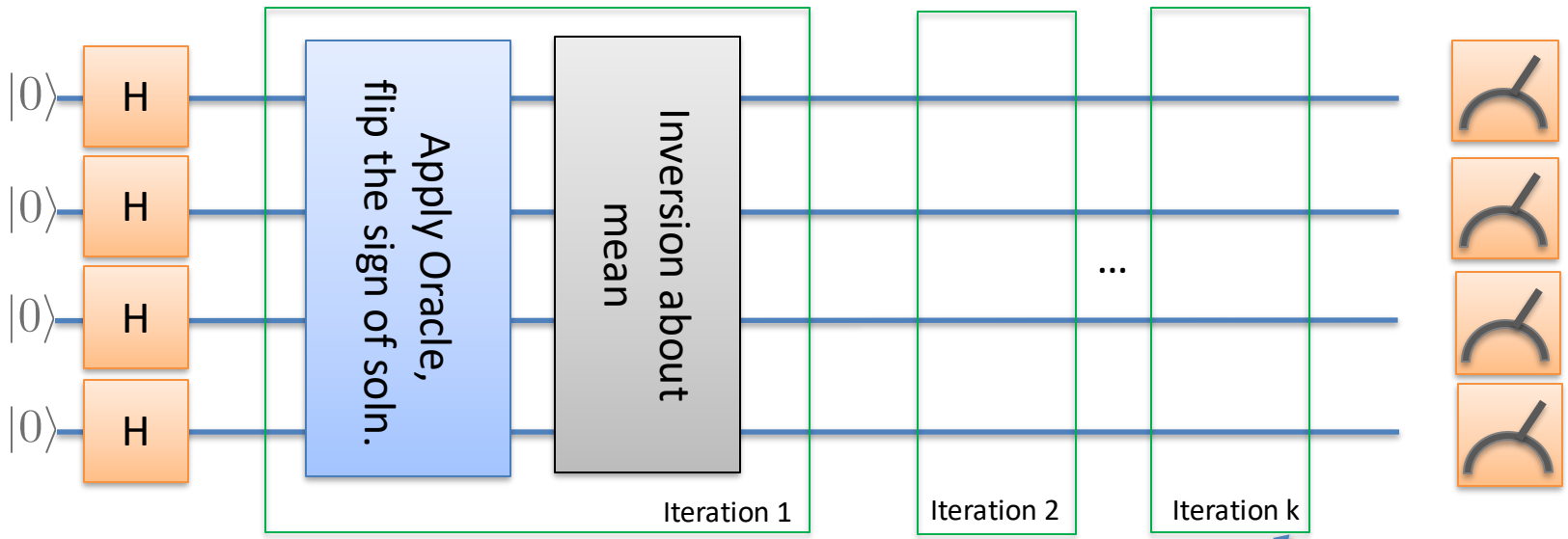
Probability of finding a solution maximum after 2 iterations of Grover step.

16.2 Probability of Success

MULT20015

Lecture 16

Number of Grover iterations

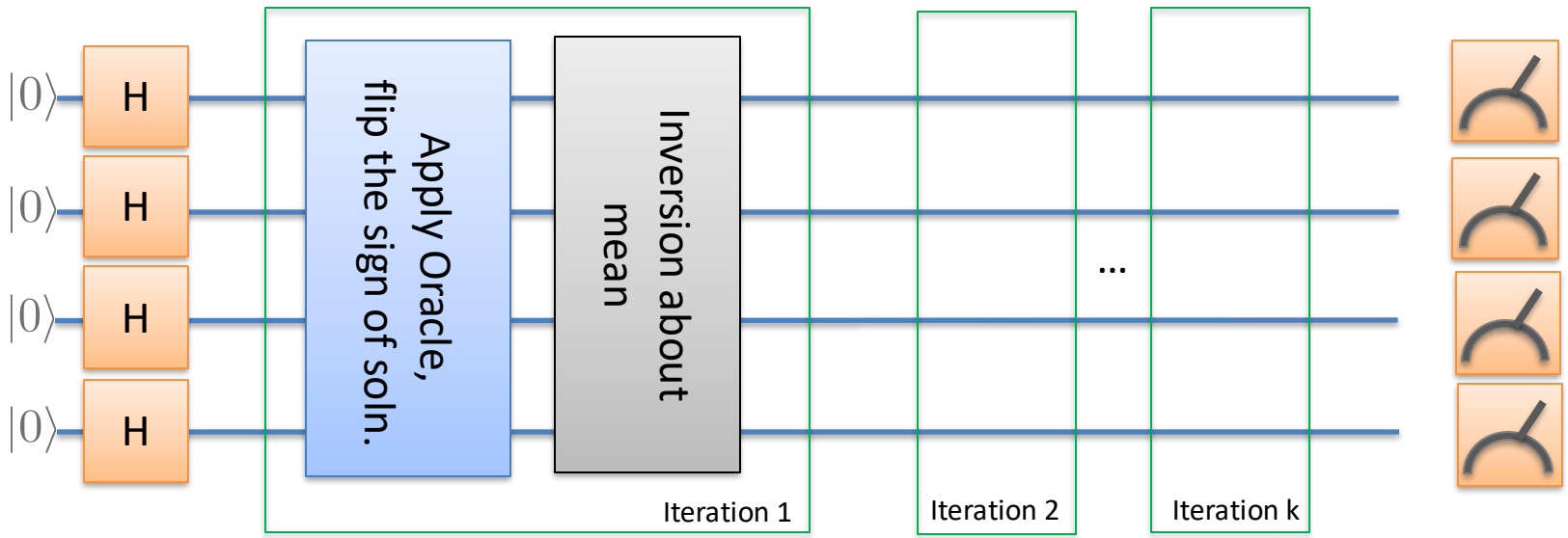


What is the optimal value of k such that amplitude of solution state is highest?

- Each iteration in Grover's algorithm performs two steps:
  - Flip the sign of the solution state
  - Inversion around the mean

- After j^{th} iteration the amplitude of solution states increase (under certain assumptions)
  - $M \ll N$, where M is the number of solutions.
  - $j \leq \sqrt{N/M}$

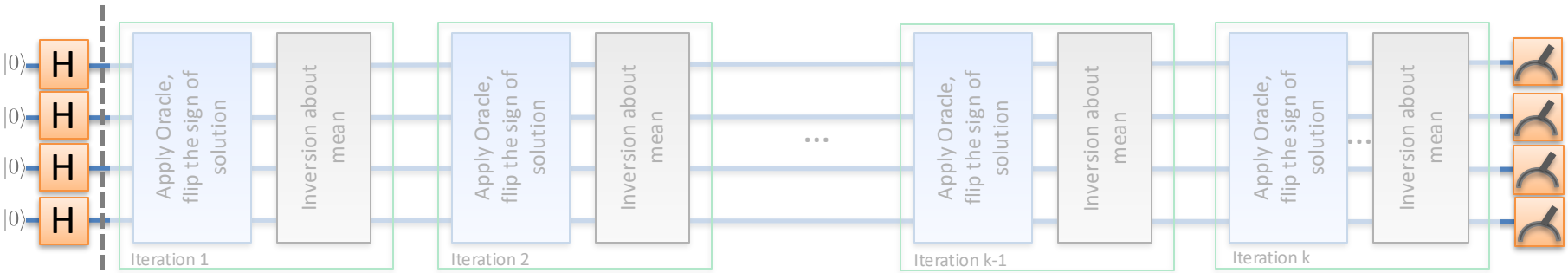
Number of Grover iterations



What is the optimal value of k such that probability t is highest?

- Alternate perspective: Let t denote the probability of picking a random value $x \in \{0,1,\ldots,N-1\}$ such that $f(x) = 1$.
- The value of t increases after j^{th} iteration (under certain assumptions)
  - $M \ll N$, where M is the number of solutions.
  - $j \leq \sqrt{N/M}$

Number of Grover iterations



$$|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$ Probability $t = \frac{M}{N} = {m_0}^2$, where M is the number of solutions.

- Let $|\psi_M\rangle = \frac{1}{\sqrt{M}} \sum_{x \in S} |x\rangle$ and $|\psi_P\rangle = \frac{1}{\sqrt{P}} \sum_{x \notin S} |x\rangle$ where
  - S is the set of solutions, M is the number of solutions, and $P = N - M$.
  - $|\psi_M\rangle$ are the states that represent solution and $|\psi_P\rangle$ represent non-solution states.

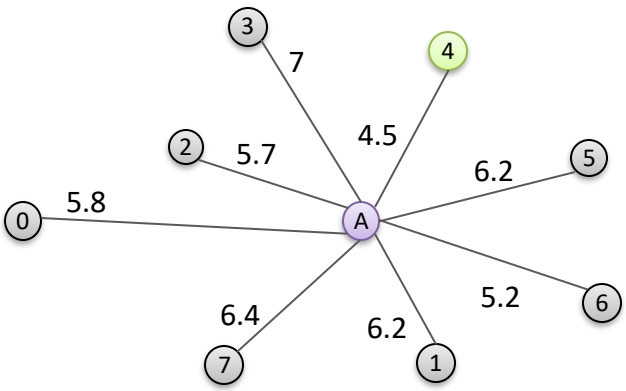
$$|\psi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle = \sqrt{\frac{M}{N}}\, |\psi_M\rangle + \sqrt{\frac{P}{N}}\, |\psi_P\rangle = m_0 |\psi_M\rangle + \mathrm{p}_0 |\psi_P\rangle$$

Each iteration changes the amplitudes.

Two basis states to understand Grover's algorithm!

Alice's coffee example

- Example: Alice wants to find a coffee shop within 5 mins of walking distance.



- $|\psi_0\rangle = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle = \frac{1}{\sqrt{8}}\sum_{x=0}^{7}|x\rangle$

$$= \frac{1}{\sqrt{8}}|0\rangle + \frac{1}{\sqrt{8}}|1\rangle + \frac{1}{\sqrt{8}}|2\rangle + \frac{1}{\sqrt{8}}|3\rangle + \frac{1}{\sqrt{8}}|4\rangle + \frac{1}{\sqrt{8}}|5\rangle + \frac{1}{\sqrt{8}}|6\rangle + \frac{1}{\sqrt{8}}|7\rangle$$

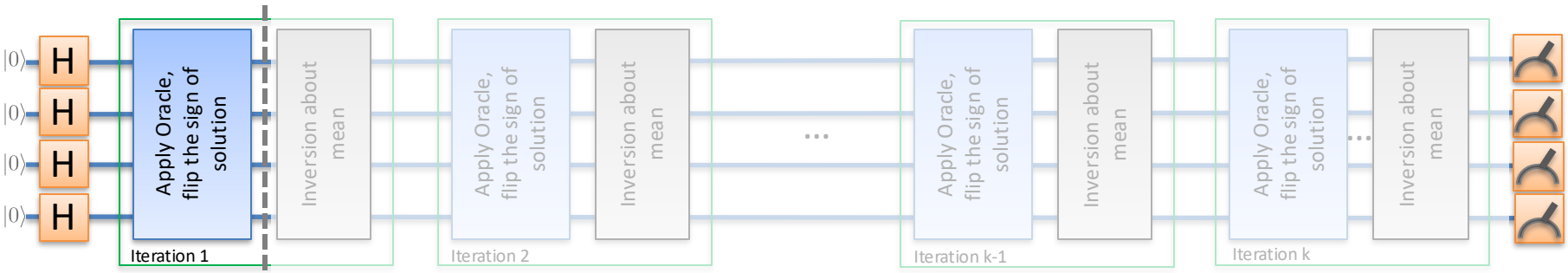
$$= \frac{1}{\sqrt{8}}|4\rangle + \frac{1}{\sqrt{8}}|0\rangle + \frac{1}{\sqrt{8}}|1\rangle + \frac{1}{\sqrt{8}}|2\rangle + \frac{1}{\sqrt{8}}|3\rangle + \frac{1}{\sqrt{8}}|5\rangle + \frac{1}{\sqrt{8}}|6\rangle + \frac{1}{\sqrt{8}}|7\rangle$$

$$= \frac{\sqrt{1}}{\sqrt{8}}|\psi_M\rangle + \frac{\sqrt{7}}{\sqrt{8}}|\psi_P\rangle$$

Re-normalise the vector components of $|\psi_P\rangle$

- Probability of finding solution $= (\frac{1}{\sqrt{8}})^2 = \frac{1}{8}$
- Average amplitude $= \frac{1}{\sqrt{8}} = 0.35355$

Number of Grover iterations

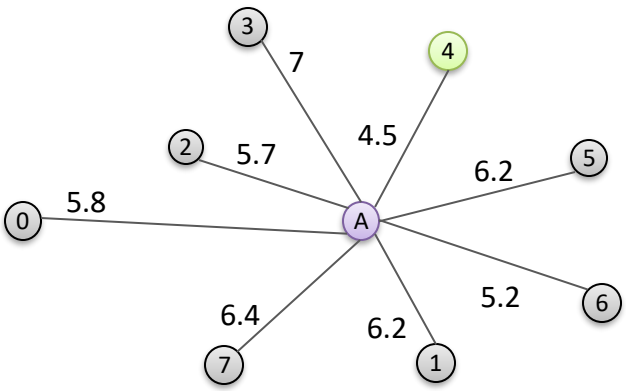


Probability $t = \frac{M}{N} = {m_0}^2$, where M is the number of solutions.

- Effect of applying the oracle and flipping the sign of solution states:
  - First iteration: O: $m_0\,|\psi_M\rangle + \mathrm{p}_0|\psi_P\rangle \rightarrow -m_0\,|\psi_M\rangle + \mathrm{p}_0|\psi_P\rangle$
  - j^{th} iteration: O: $m_j\,|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle \rightarrow -m_j\,|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle$
- Average amplitude:
  - $A_j = \frac{-\sqrt{M}m_j + \sqrt{P}p_j}{N} = \frac{-\sqrt{M}m_j + \sqrt{N-M}p_j}{N}$
- Note: After applying the oracle, the amplitude changes sign but the probability t does not change.

Alice's coffee example

- Example: Alice wants to find a coffee shop within 5 mins of walking distance.



- $O: |\psi_0\rangle$

$$= -\frac{1}{\sqrt{8}}|4\rangle + \frac{1}{\sqrt{8}}|0\rangle + \frac{1}{\sqrt{8}}|1\rangle + \frac{1}{\sqrt{8}}|2\rangle + \frac{1}{\sqrt{8}}|3\rangle + \frac{1}{\sqrt{8}}|5\rangle + \frac{1}{\sqrt{8}}|6\rangle + \frac{1}{\sqrt{8}}|7\rangle$$

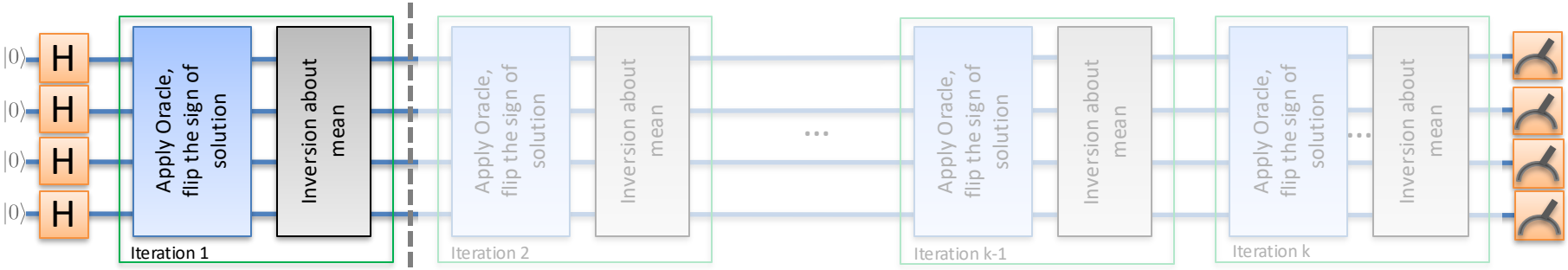
$$= -\frac{1}{\sqrt{8}}|\psi_M\rangle + \frac{\sqrt{7}}{\sqrt{8}}|\psi_P\rangle = -m_0|\psi_M\rangle + \mathrm{p}_0|\psi_P$$

- Probability of finding solution $= (-\frac{1}{\sqrt{8}})^2 = \frac{1}{8} = 0.125$

- Average amplitude $\mathrm{A} = \frac{\frac{-1}{\sqrt{8}} + 7\frac{1}{\sqrt{8}}}{8} = 0.265165$

Average amplitude is calculated on original state.

Number of Grover iterations



Probability $t = m^2{}_{j+1} = sin^2\big((2j+1)\theta\big)$,

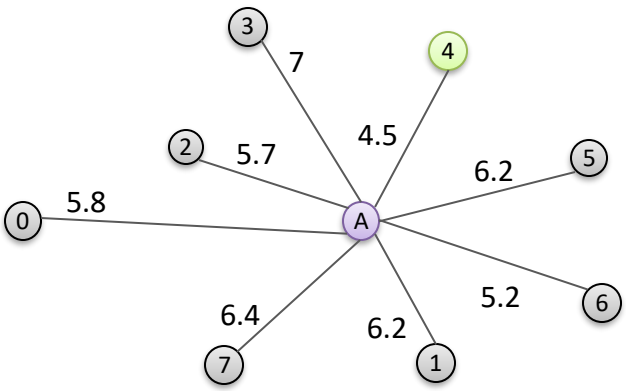
where j is the number of iterations and $\sin(\theta) = \sqrt{M/N}$

- Inversion about the mean transforms:
  - j^{th} iteration: $-m_j|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle \rightarrow (2A_j\sqrt{M} + m_j)|\psi_M\rangle + (2A_j\sqrt{N-M} - p_j)|\psi_P\rangle$
    $= m_{j+1}|\psi_M\rangle + \mathrm{p}_{\mathrm{j+1}}|\psi_P\rangle$

- Solution to the recurrence equations:
  - $m_{j+1} = \sin\big((2j+1)\theta\big)$
  - $p_{j+1} = \cos\big((2j+1)\theta\big)$
  - $\sin(\theta) = \sqrt{\frac{M}{N}}$

Alice's coffee example

- Example: Alice wants to find a coffee shop within 5 mins of walking distance.



- $|\psi_1\rangle = \sum_{j=0}^{N-1}(2A - a_j)|x_j\rangle$

$$= \left(2A + \frac{1}{\sqrt{8}}\right)|4\rangle + (2A - \frac{1}{\sqrt{8}})|0\rangle + (2A - \frac{1}{\sqrt{8}})|1\rangle + (2A - \frac{1}{\sqrt{8}})|2\rangle + (2A - \frac{1}{\sqrt{8}})|3\rangle + (2A - \frac{1}{\sqrt{8}})|5\rangle + (2A - \frac{1}{\sqrt{8}})|6\rangle + (2A - \frac{1}{\sqrt{8}})|7\rangle$$

$$= \left(2A + \frac{1}{\sqrt{8}}\right)|\psi_M\rangle + \left(2A\sqrt{7} - \frac{\sqrt{7}}{\sqrt{8}}\right)|\psi_P\rangle$$

- Probability of finding solution $= (2A + \frac{1}{\sqrt{8}})^2 = (2*0.265165 + \frac{1}{\sqrt{8}})^2 = 0.781249$

How many iterations required?

Case: Single solution

$$\sin\theta = \frac{1}{\sqrt{N}}$$

For small angles,

$$\theta \approx \frac{1}{\sqrt{N}}$$

After k iterations, we rotate to have only marked solutions:

$$(2k+1)\theta = \frac{\pi}{2}$$

$$k \approx \frac{\pi}{4}\sqrt{N}$$

The number of steps, k, required scales as O(√N), and not with N as it would classically.

This is a “polynomial” rather than an “exponential” speedup.

How many iterations required?

Case: Multiple solutions

$$\sin\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

For small angles,

$$\theta \approx \frac{\sqrt{M}}{\sqrt{N}}$$

After k iterations, we rotate to have only marked solutions:

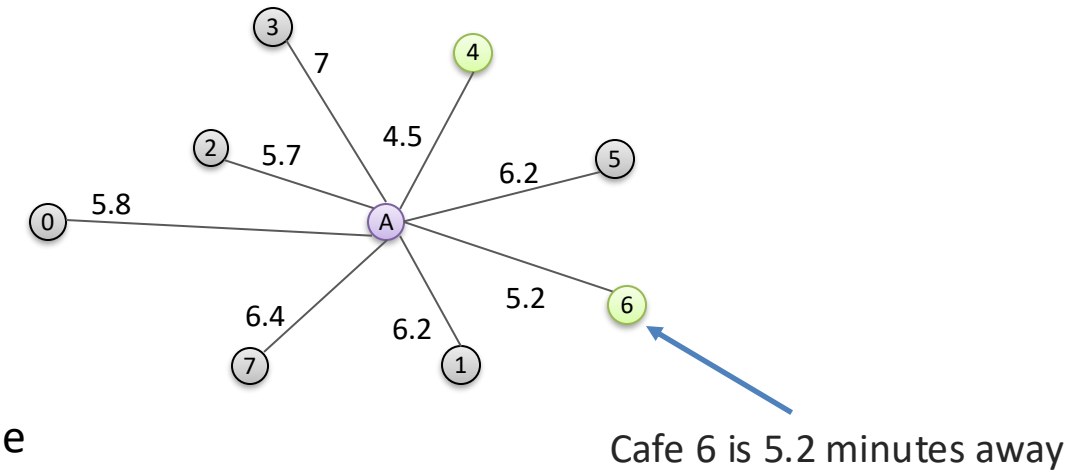
$$(2k+1)\theta = \frac{\pi}{2}$$

$$k \approx \frac{\pi}{4}\sqrt{\frac{N}{M}}$$

Having multiple solutions is faster than searching for a single marked solution.

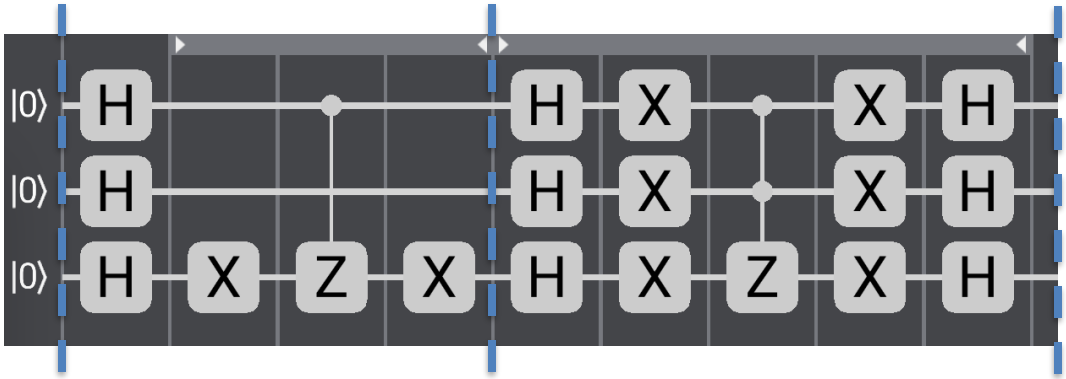
Alice's coffee example: Many solutions

- Example: Alice wants to find a coffee shop within 5.5 mins of walking distance.

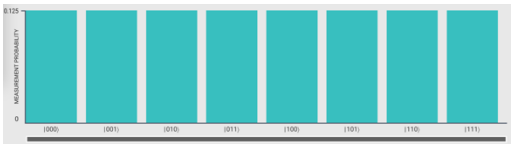


- $X = \{0,1,\dots,7\}$
  - X is not ordered by travel time
- $f(x) = \begin{cases} 1, \text{ if time(x)} \leq 5.5 \\ 0, \text{otherwise} \end{cases}$
  - In general, the function f does not need to know about the search space.
- Solution $x \in \{4, 6\}$

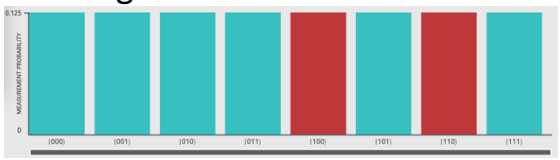
Alice's coffee example: Many solutions



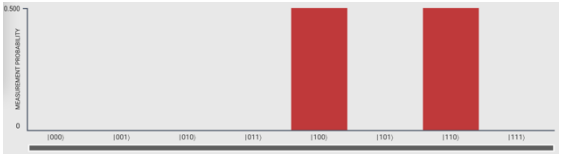
Equal superposition



Marking of the solution state



Mean inversion



0.5 0.5

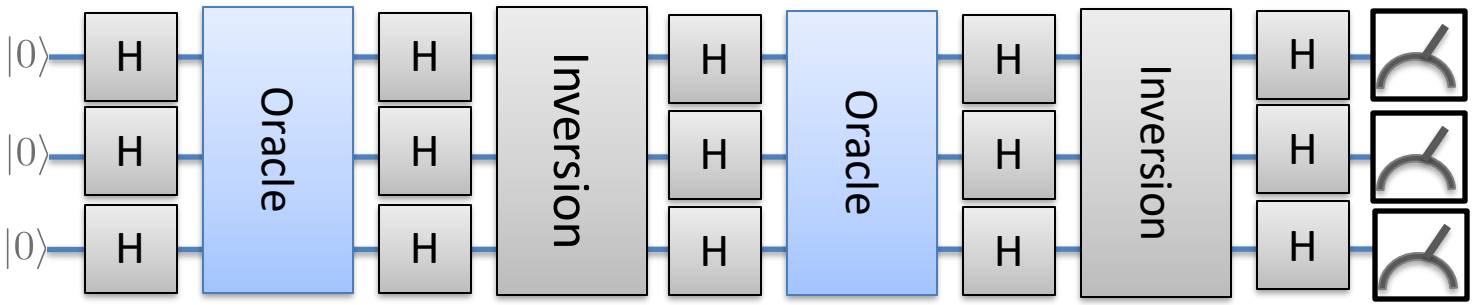
Probability of finding a solution maximum after 1 iterations of Grover step.

16.3 Geometric Interpretation

MULT20015

Lecture 16

Geometric interpretation of Grover's algorithm



Recall:

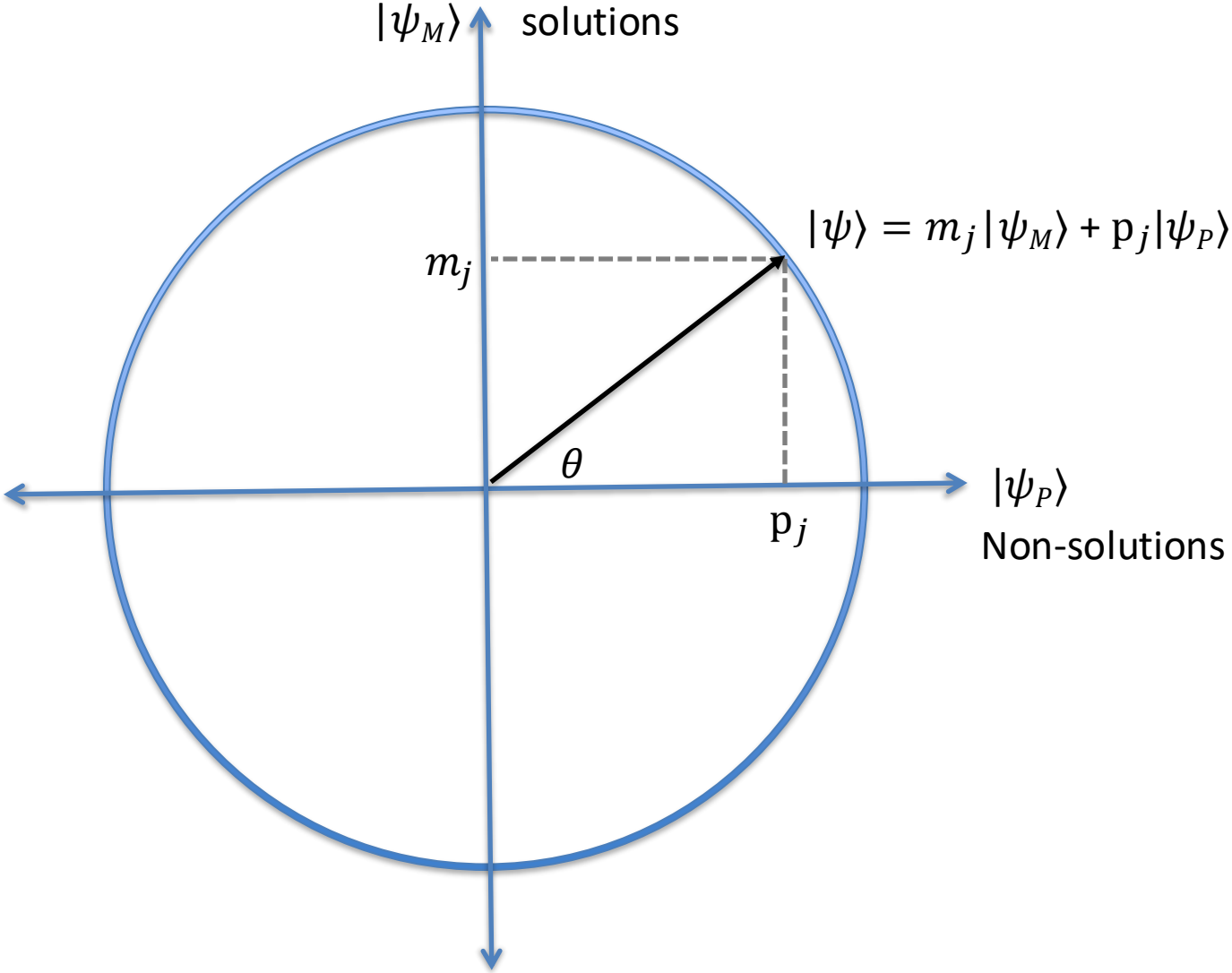
$$|\psi_0\rangle = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle = \sqrt{\frac{M}{N}}\,|\psi_M\rangle + \sqrt{\frac{P}{N}}\,|\psi_P\rangle = m_0\,|\psi_M\rangle + \mathrm{p}_0|\psi_P\rangle$$

Solutions!

Non-solutions

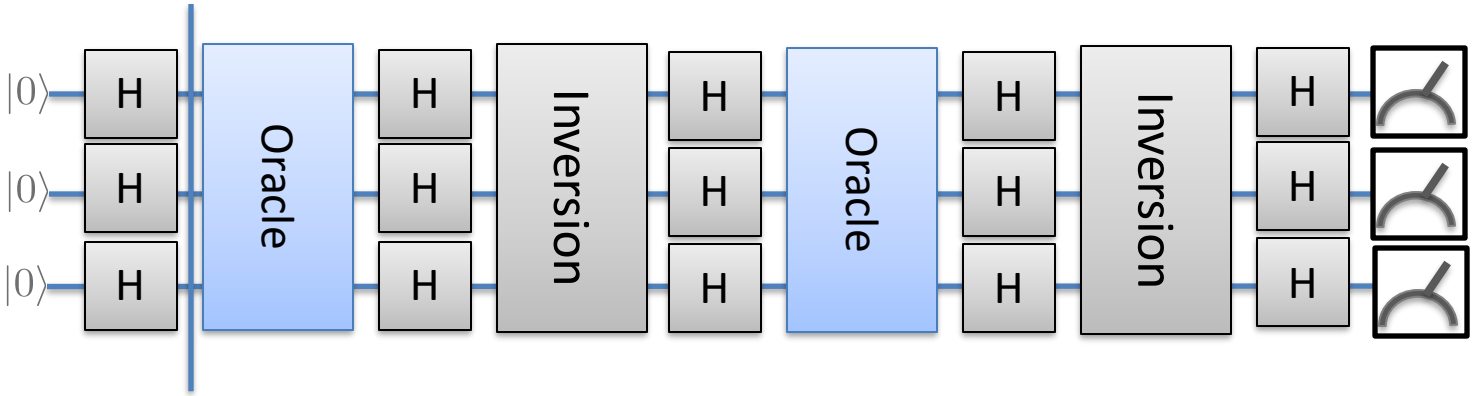
We only need to consider the amplitude of these two states in Grover's algorithm. Every operation is also real, so we can plot on a circle.

Geometric Interpretation



Every state in Grover's algorithm can be expressed as a superposition of these vectors

Equal superposition



Equal superposition state:

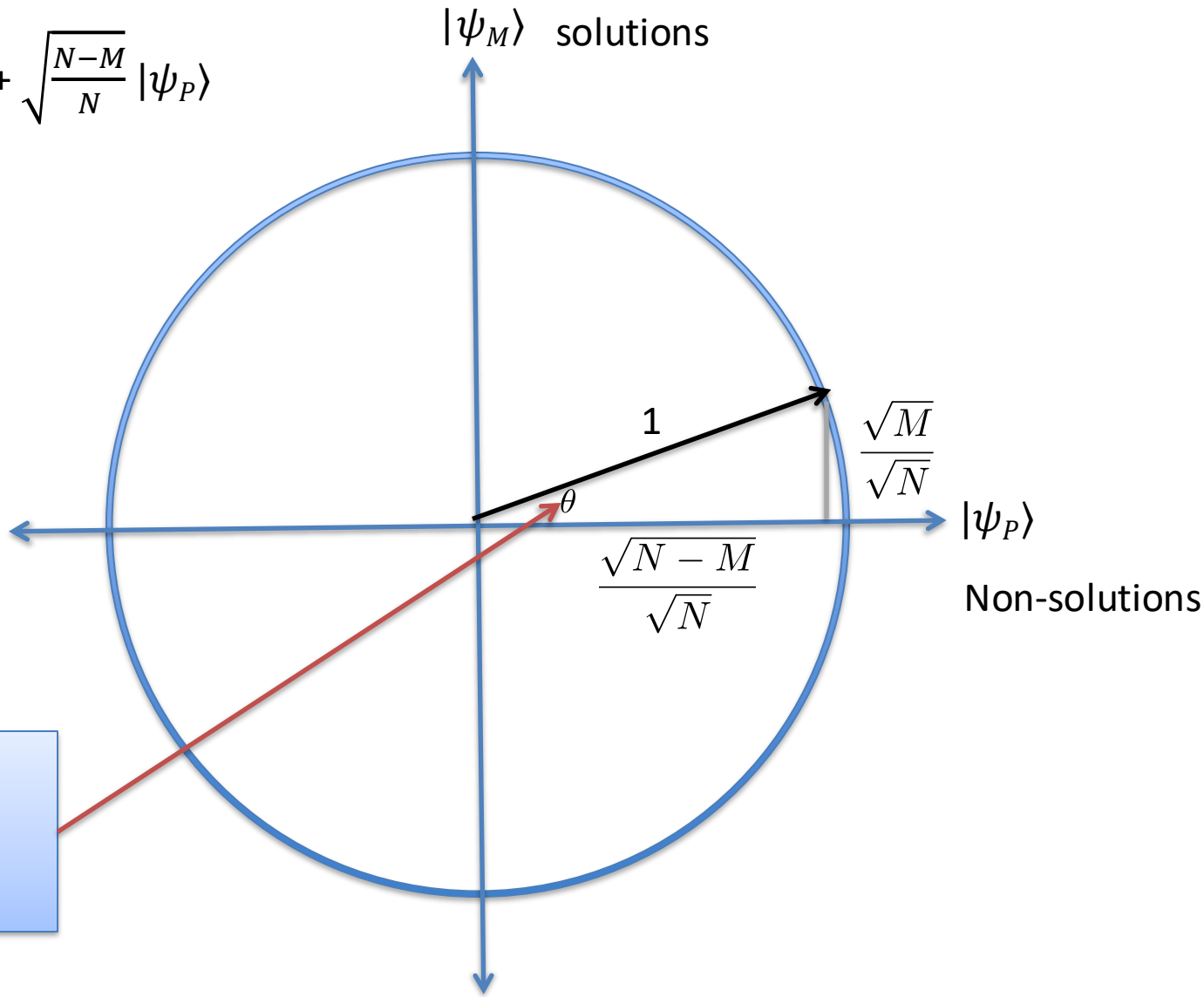
$$|\psi_0\rangle = \frac{1}{\sqrt{N}}\sum_{x=0}^{N-1}|x\rangle = \sqrt{\frac{M}{N}}\,|\psi_M\rangle + \sqrt{\frac{P}{N}}\,|\psi_P\rangle$$

$$= \sqrt{\frac{M}{N}}\,|\psi_M\rangle + \sqrt{\frac{N-M}{N}}\,|\psi_P\rangle$$

Equal Superposition

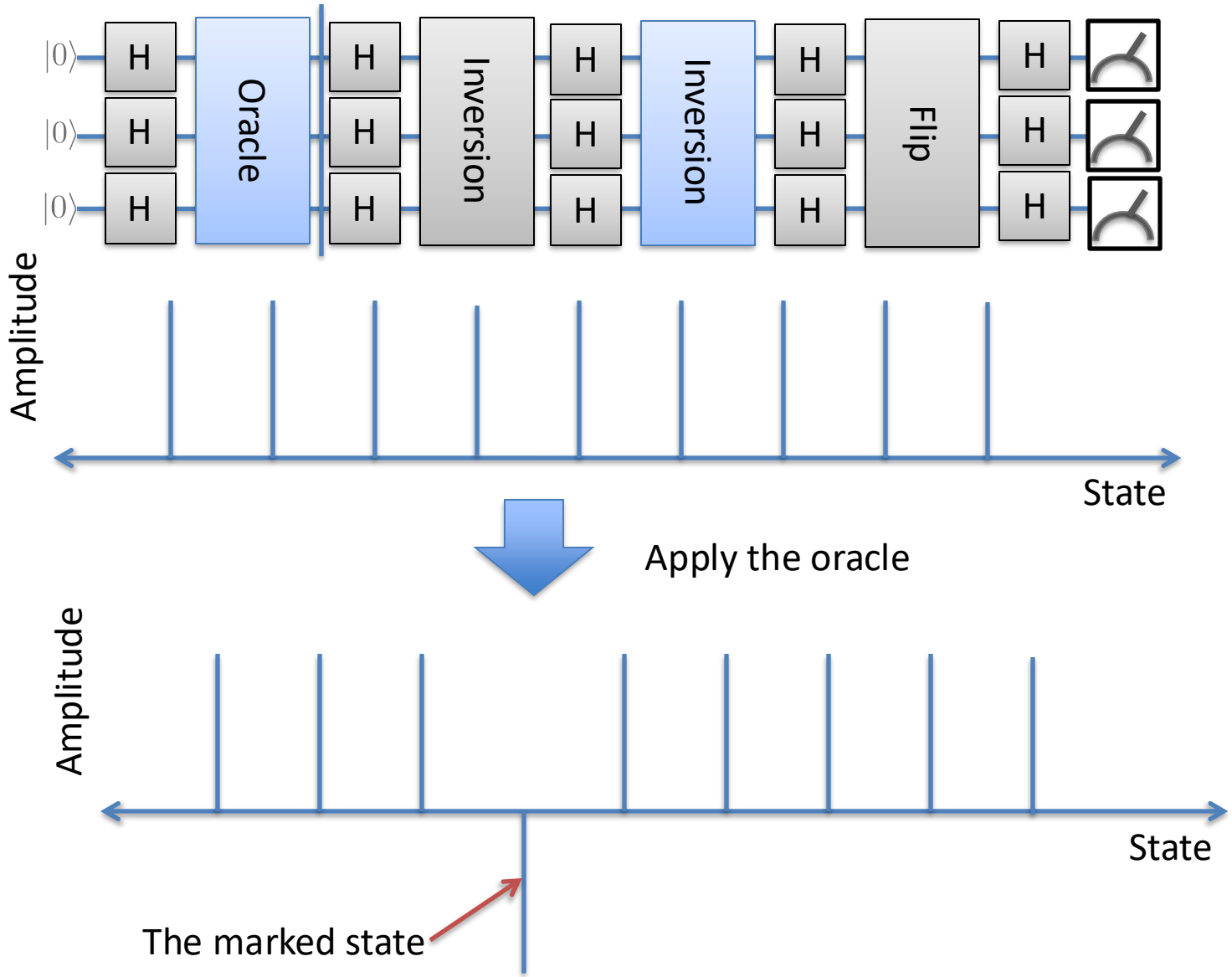
Consider the equal superposition:

$$|\psi_0\rangle = \sqrt{\frac{M}{N}}\,|\psi_M\rangle + \sqrt{\frac{N-M}{N}}\,|\psi_P\rangle$$



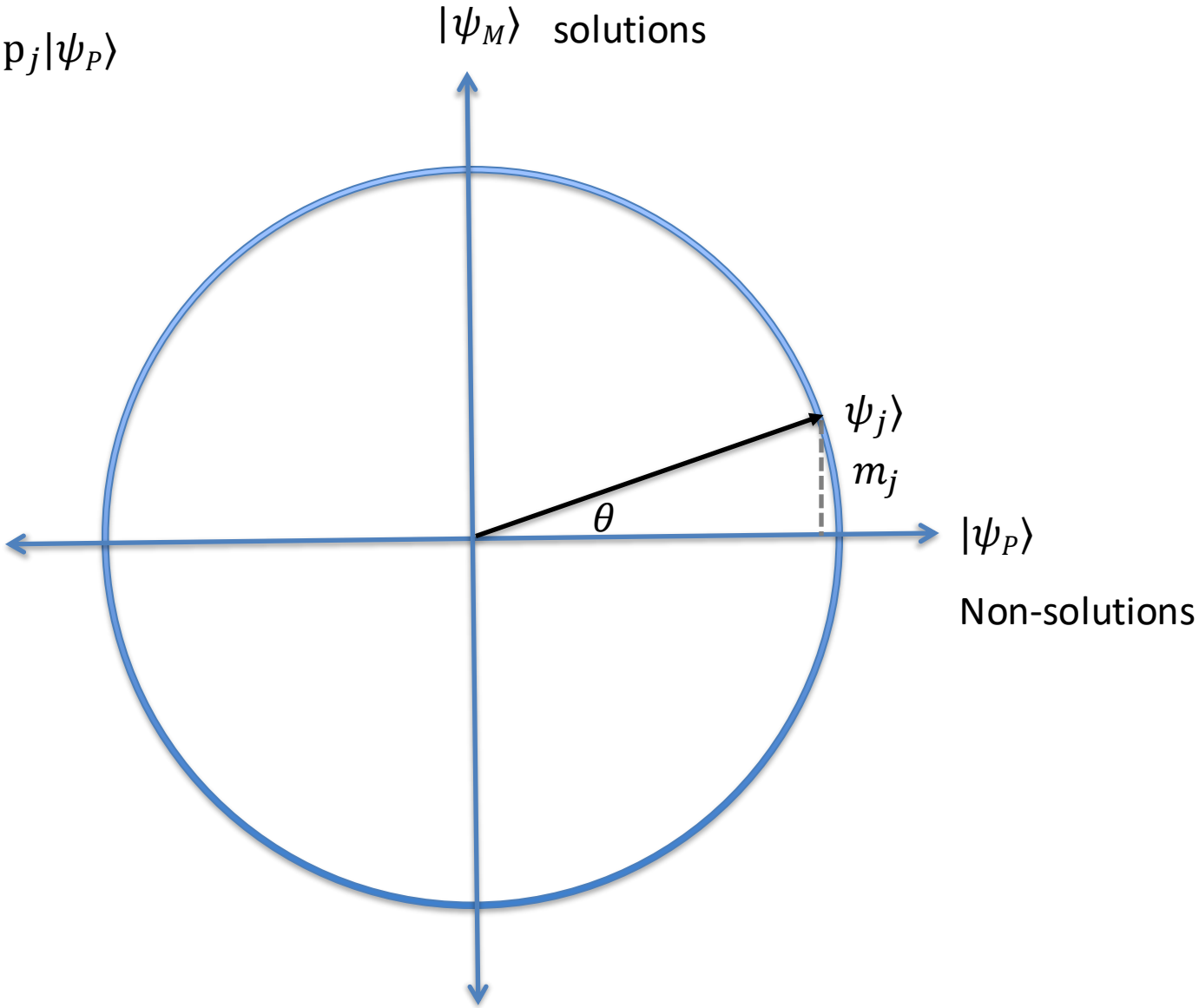
$$\sin\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

Effect of the Oracle



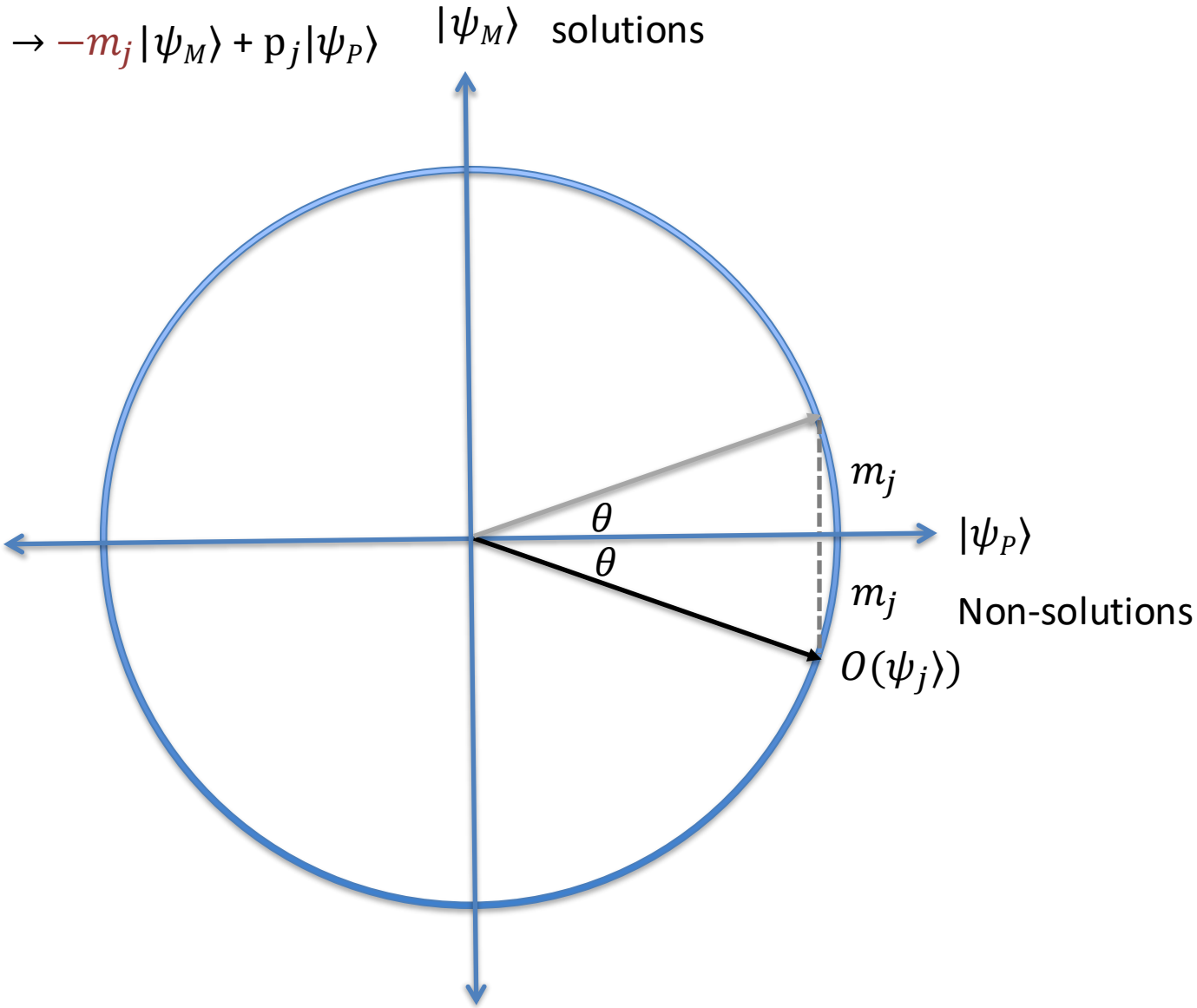
Geometric Effect of Oracle

$$|\psi_j\rangle = m_j|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle$$

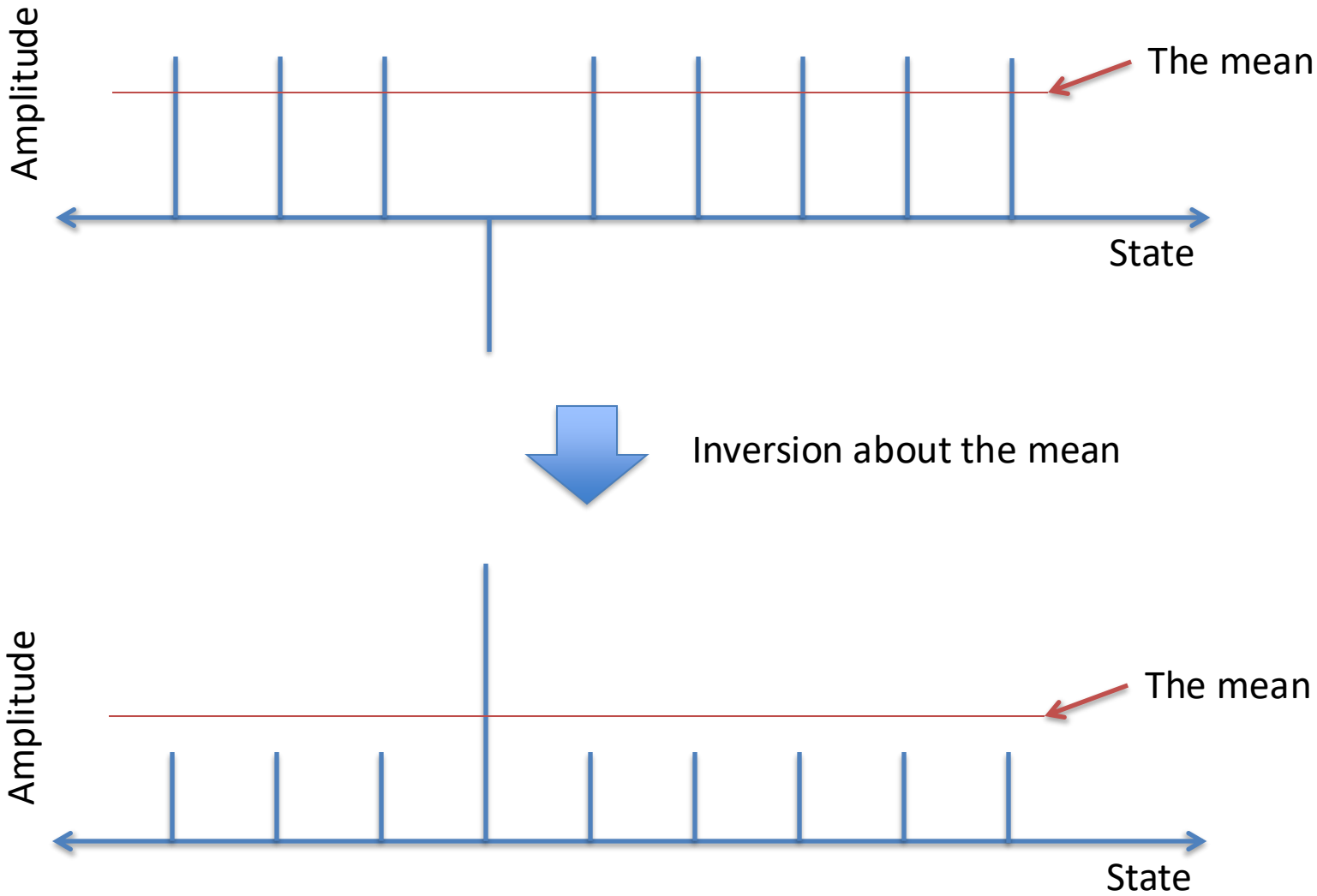


Geometric Effect of Oracle

$O: m_j\,|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle \rightarrow -m_j\,|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle$

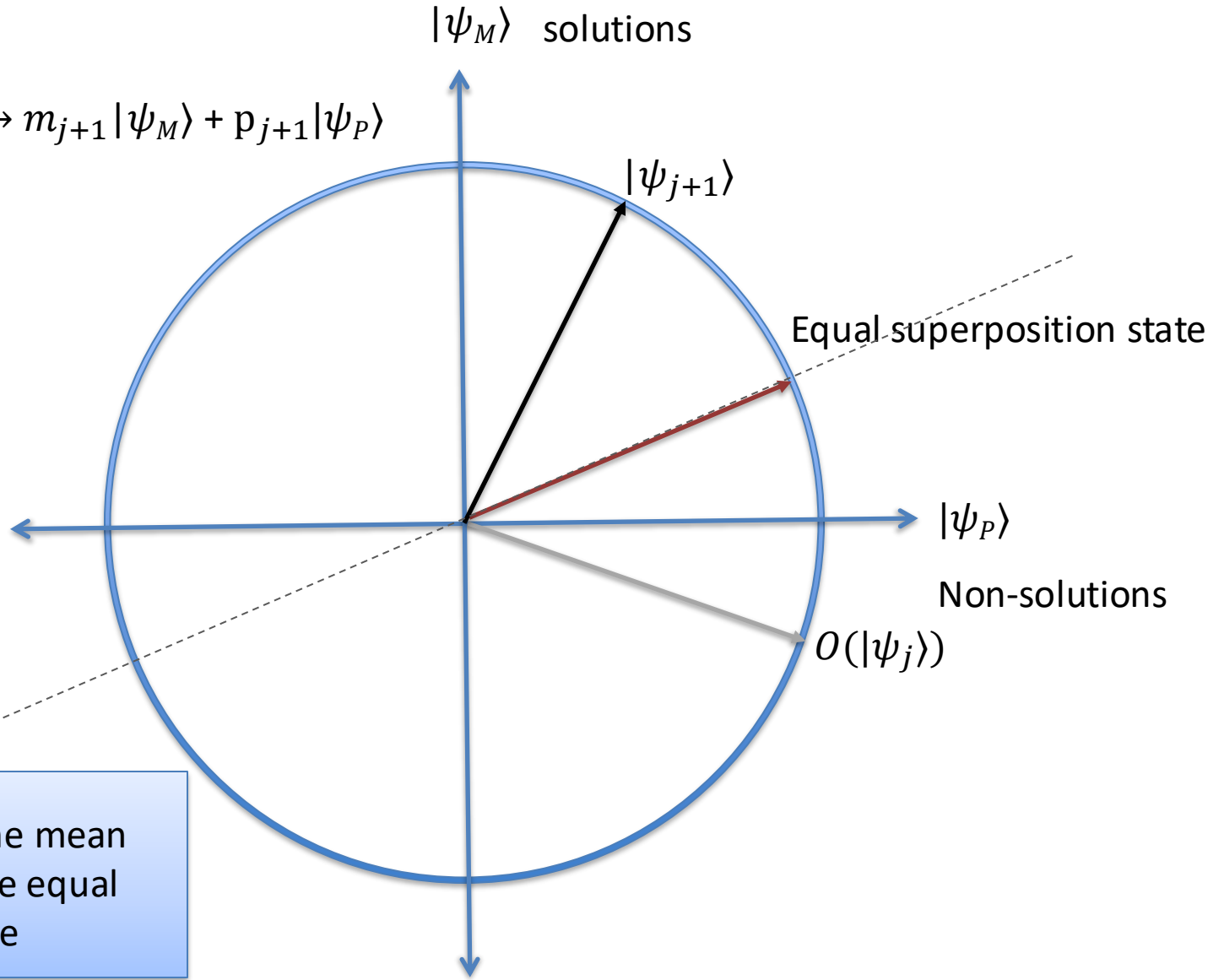


Effect of inversion about the mean



Geometric Effect of Oracle

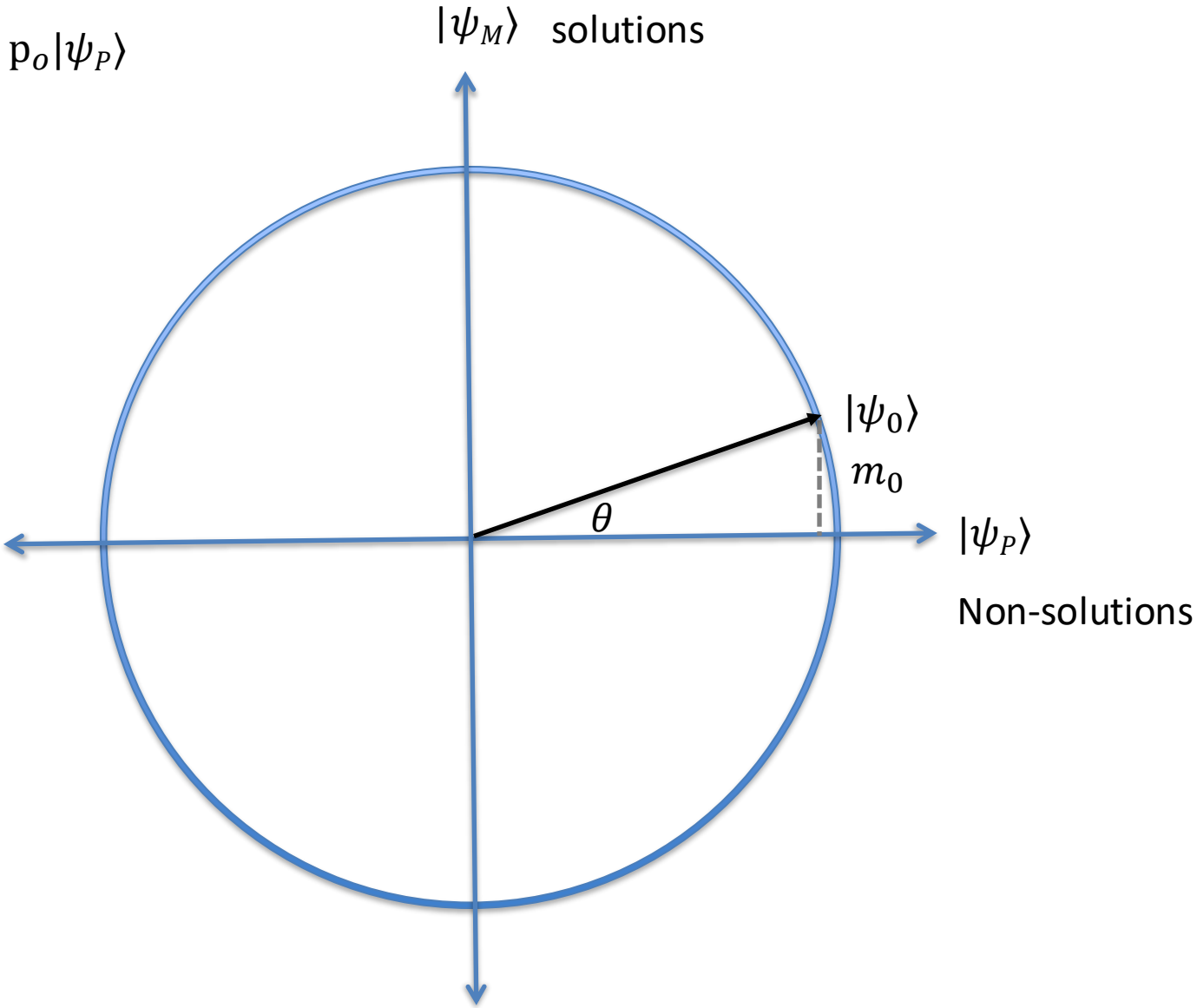
$$-m_j|\psi_M\rangle + \mathrm{p}_j|\psi_P\rangle \rightarrow m_{j+1}|\psi_M\rangle + \mathrm{p}_{j+1}|\psi_P\rangle$$



Inversion about the mean reflects around the equal superposition state

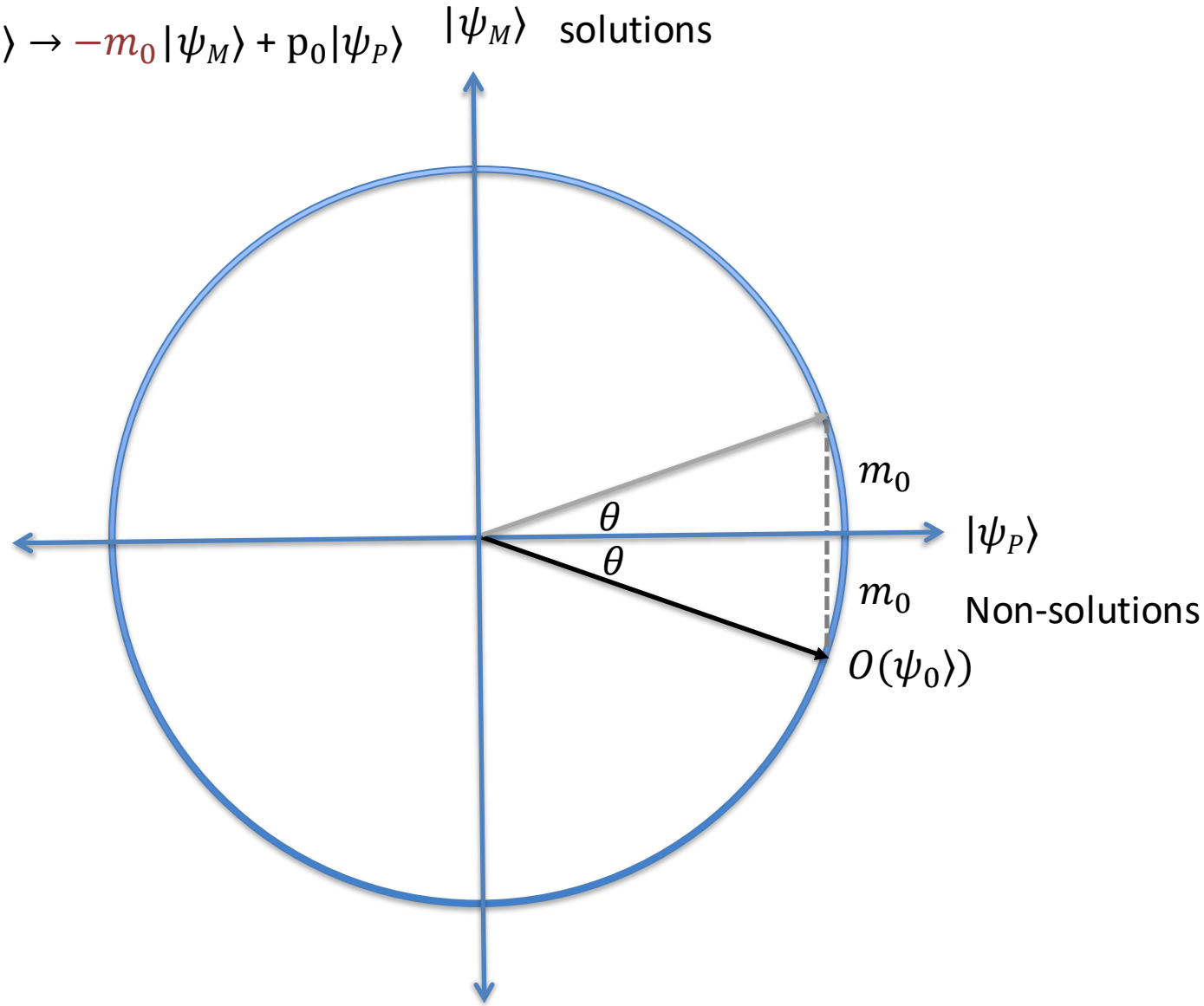
Geometric Effect of Both Oracle and Inversion

$$|\psi_0\rangle = m_o|\psi_M\rangle + \mathrm{p}_o|\psi_P\rangle$$



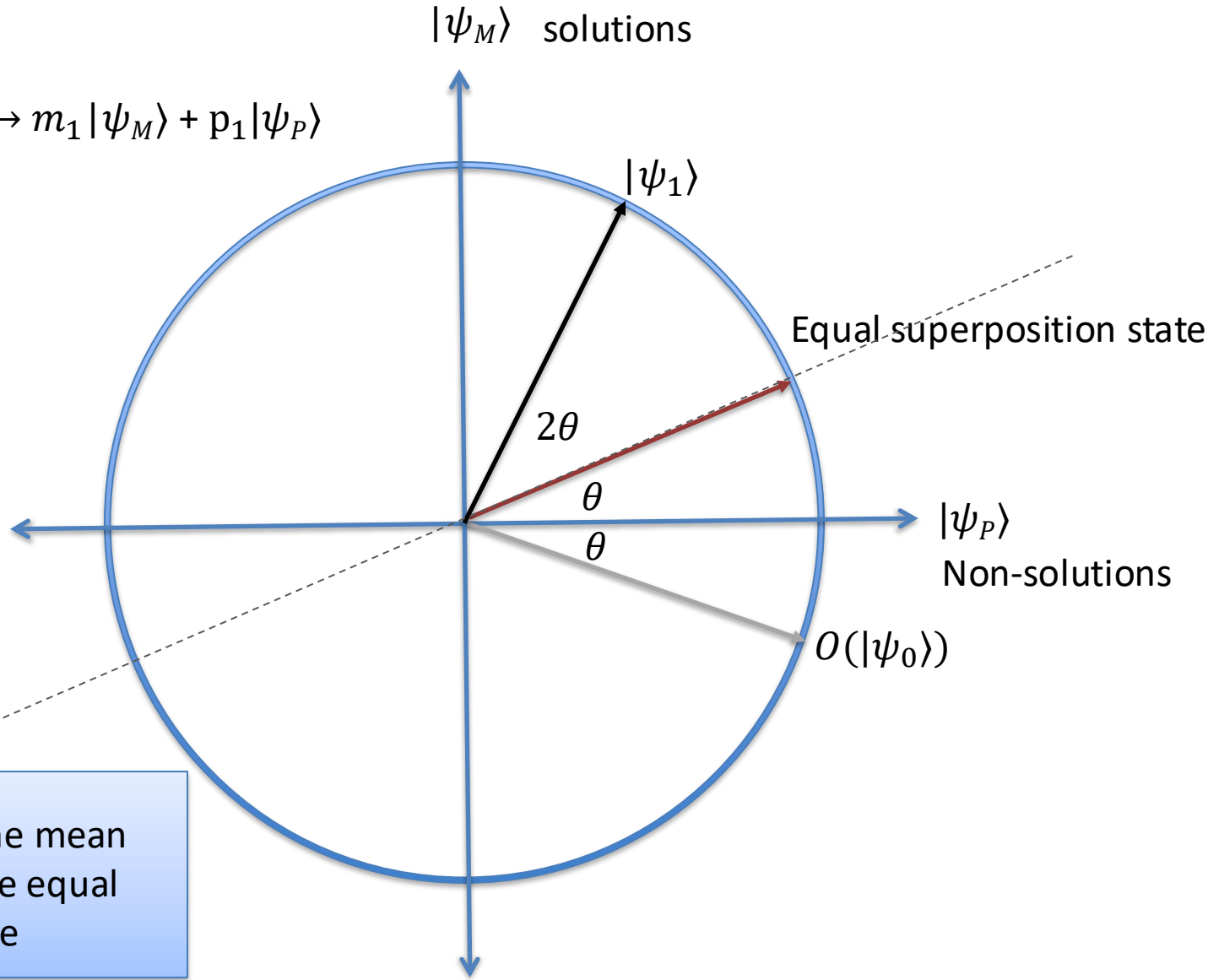
Geometric Effect of Both Oracle and Inversion

$O: m_0|\psi_M\rangle + \mathrm{p}_0|\psi_P\rangle \rightarrow -m_0|\psi_M\rangle + \mathrm{p}_0|\psi_P\rangle$



Geometric Effect of Both Oracle and Inversion

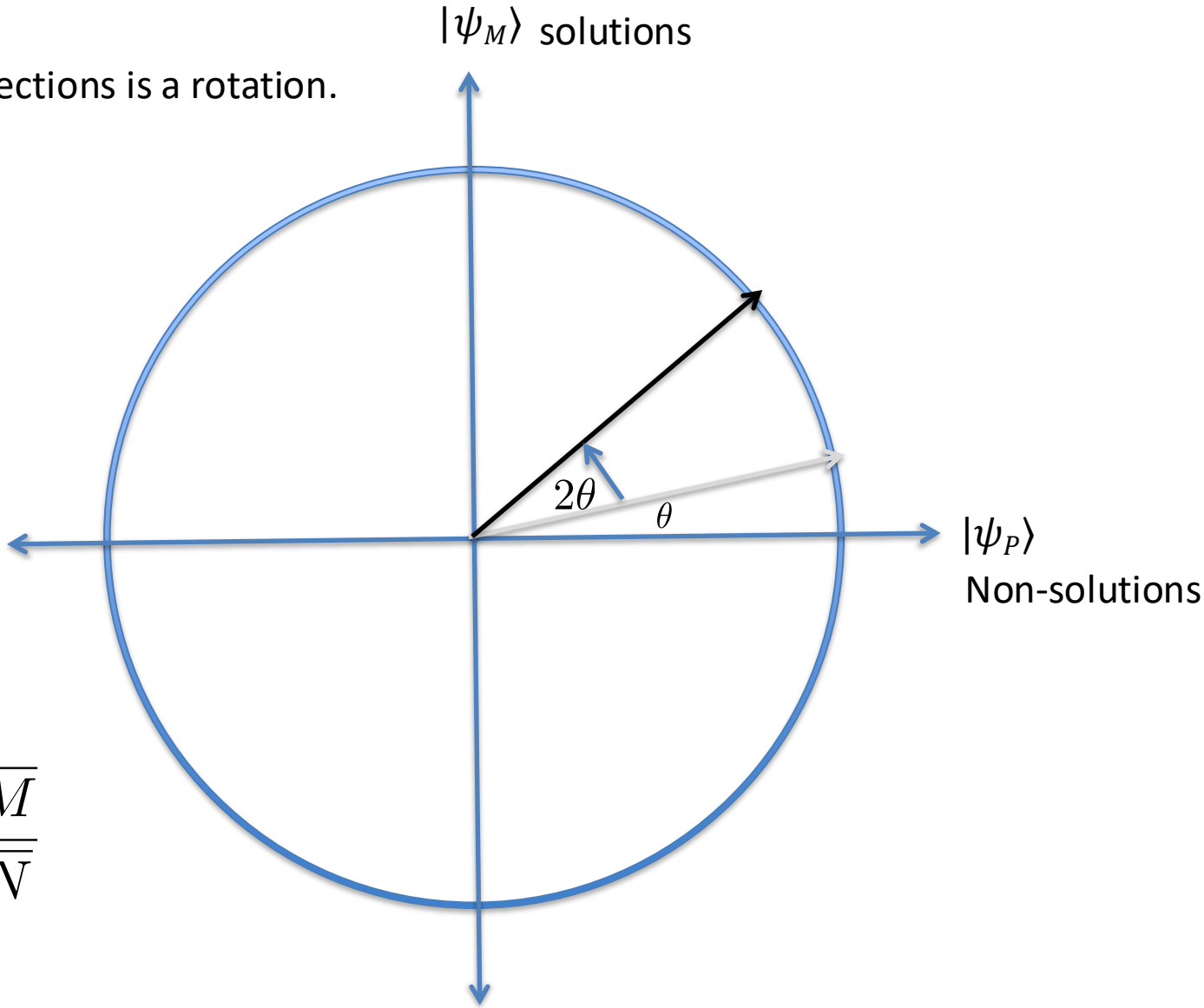
$$-m_0|\psi_M\rangle + \mathrm{p}_0|\psi_P\rangle \rightarrow m_1|\psi_M\rangle + \mathrm{p}_1|\psi_P\rangle$$



Inversion about the mean reflects around the equal superposition state

Total effect of one Grover iteration

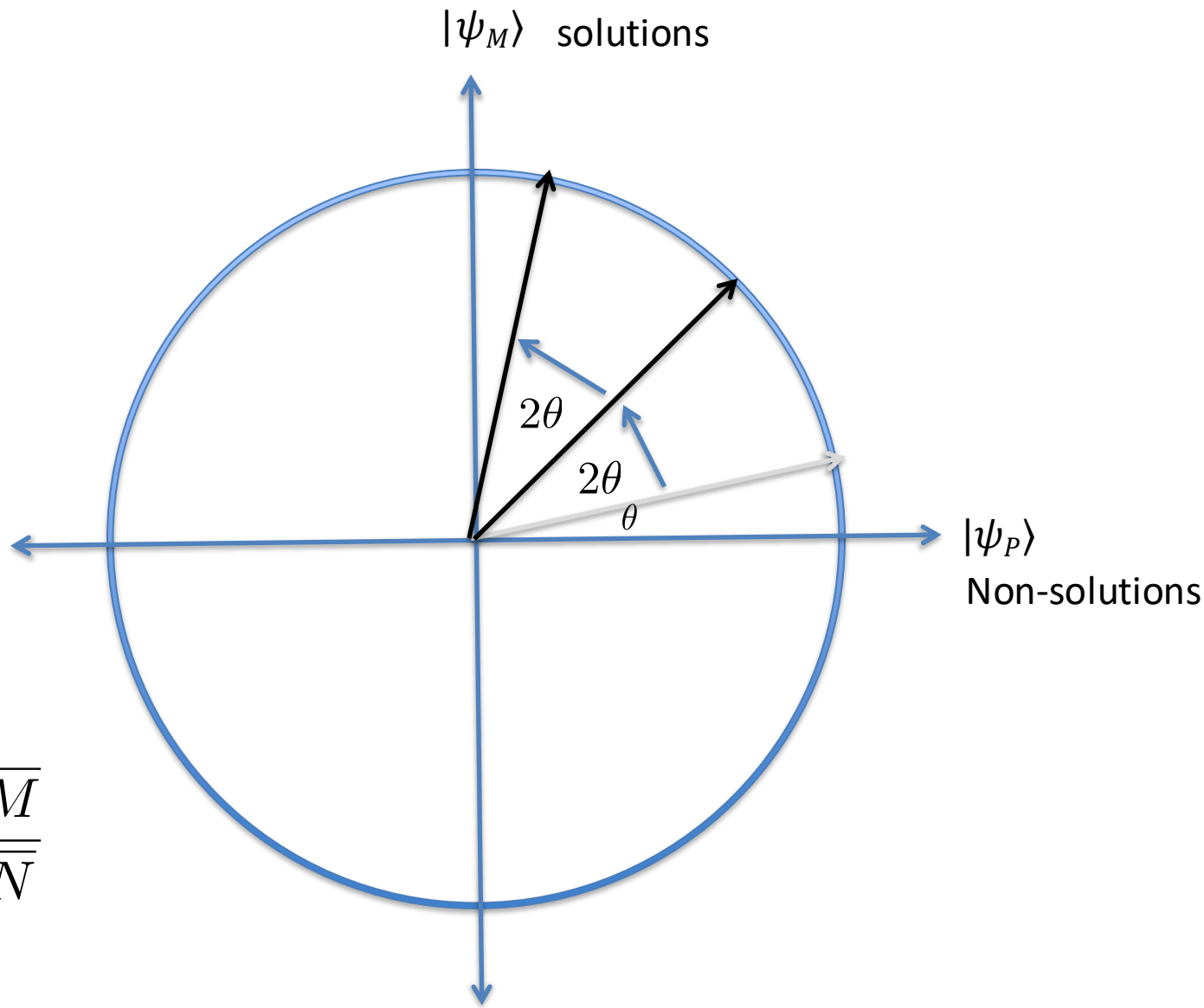
Product of two reflections is a rotation.



$$\sin\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

Many Grover iterations

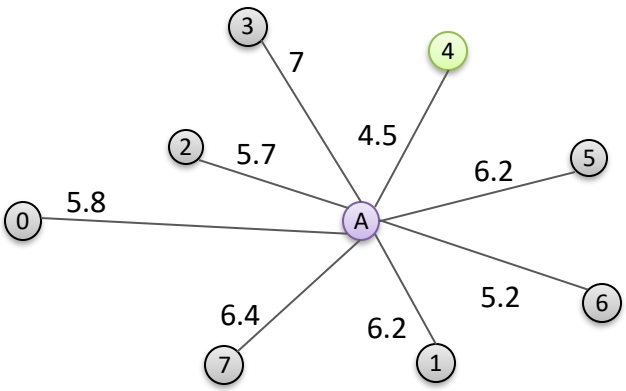
Product of two reflections is a rotation.



$$\sin\theta = \frac{\sqrt{M}}{\sqrt{N}}$$

Alice's coffee example

- Example: Alice wants to find a coffee shop within 5 mins of walking distance.

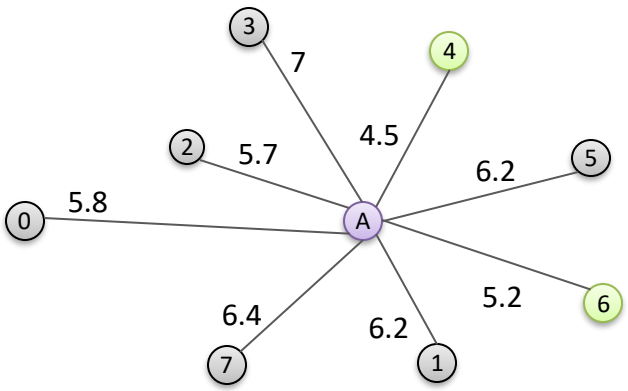


- $\sin(\theta) = \frac{1}{\sqrt{8}}$ or $\theta = 0.361367$rad
- Probability after iteration $j = \sin^2\big((2j+1)\theta\big)$
- Probability of soln. after iteration 0= $\sin^2(\theta) = \sin^2(0.361367) = 0.125$
- Probability of soln. after iteration 1 = $\sin^2(3\theta) = \sin^2(1.084101) = 0.781249$
- Probability of soln. after iteration 2 = $\sin^2(5\theta) = \sin^2(1.80635) = 0.945312$

Ex. What happens after one more iteration?

Alice's coffee example with multiple solutions

- Example: Alice wants to find a coffee shop within 5.5 mins of walking distance.



- $\sin(\theta) = \frac{\sqrt{2}}{\sqrt{8}}$ or $\theta = 0.523598$rad
- Probability after iteration $j = \sin^2\big((2j+1)\theta\big)$
- Probability of soln. after iteration 0= $\sin^2(\theta) = \sin^2(0.523598) = 0.25$
- Probability of soln. after iteration 1 = $\sin^2(3\theta) = \sin^2(1.5707) = 1$

Ex. What happens after one more iteration?

Week 8

Lecture 15

15.1 Grover's algorithm: Intuition

15.2 The Oracle

15.3 Inversion about the mean

15.4 Example Quantum Search

Lecture 16

16.1 Grover's algorithm Example

16.2 Probability of success

16.3 Geometric interpretation

Practice class 8

Grover's algorithm using QUI

Subject outline

Lecture topics (by week)

1 – Introduction to quantum computing and maths basics
2 – Single qubit representations and logic operations
3 – Two qubit states and logic gates
4 – Multi-qubit states and quantum arithmetic
5 – Basic quantum algorithms
6 – Period finding, cryptography and quantum factoring
7 – Shor's algorithm, post-quantum crypto, quantum key distribution
8 – Quantum search algorithms
9 – Grover search applications, optimisation problems
10 – Solving optimisation problems on quantum computers
11 – Applications in quantum machine learning
12 – Real quantum computer devices

Assignment schedule:
#1: Hand out in Week 2
#2: Hand out in Week 8